Pilot Overview:
Numina Sensor Measurement of Multimodal Activity and Shared Streets

SUMMARY & BACKGROUND

This project began with the announcement of Boston's "Healthy Streets" program which proposed accelerating several planned changes to Boston's streets to support a healthy reopening and an equitable recovery. In order to facilitate a safe re-opening and abide by social distancing guidelines in the context of Covid-19, the City of Boston used creative strategies to more quickly realize long-term mobility goals of equity, economic opportunity, and climate responsiveness. Temporary changes to Boston's streets were implemented to support a healthy reopening and an equitable recovery, and the City required a quantitative measurement solution in order to evaluate the effectiveness and desirability of these changes.

One of the primary values guiding "Healthy Streets" was safety in the context of increasing safety for bicyclists and pedestrians. This value begged the question: What makes a street healthy for everyone, particularly in the context of technology which captures data about the public realm? For many, surveillance technology and invasive data collection directly contributes to a decreased feeling of health and safety.

Always seeking to protect the privacy and security of its constituents, the City strives to collect data in the public realm using measurement solutions which are transparent and secure, while promoting open data practices and gathering the minimum data required for the stated task. So, in the summer of 2020 the Mayor's Office of New Urban Mechanics (MONUM) embarked on a six-month pilot of Numina sensors to measure the movement patterns and volume of bikes, cars, trucks and pedestrians at three distinct intersections.

Today, much of the data collected for transportation planning purposes is generated either manually or from camera footage, which can involve storing visually-identifiable images of faces and license plates. Once collected, this information is often proprietary and users such as city planners must pay regular fees to access it. Rarely does the information become publicly available and accessible to a broader audience. By contrast, this project's dataset was generated without the collection or storage of sensitive images and will be freely available to the public for download in multiple formats and through a variety of platforms.

Description of Pilot

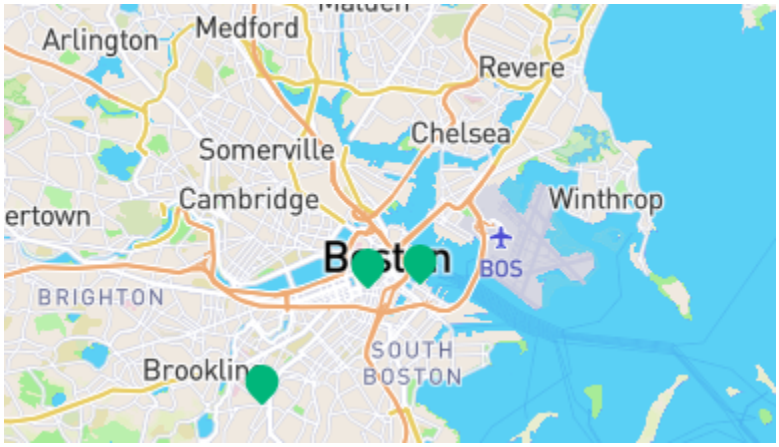
The primary purpose of this pilot was to understand the impacts of temporary street-level changes that would be implemented to facilitate a safe re-opening in the context of Covid-19. A secondary objective of the project was to evaluate a privacy-oriented solution to data collection in the public realm.

Sensors were installed at three distinct locations:

- In the Seaport district on a commercial street with bike lanes (Northern Avenue)
- Downtown at a busy intersection next to the Boston Common (Tremont Street)

- And in Jamaica Plain where the southwest corridor convenes with a blue bike station and T stop (Jackson Sq.)

Pilot Project Area



Sensor locations:

- **Tremont St & Boylston St,** Boston, MA 02116, 42.352371214962034, -71.06449848225252
- **254 Centre St,** Boston, MA 02130, 42.322802, -71.100588
- **50 Northern Ave**, Boston, MA 02210, 42.35362972361076, -71.04612808140988

I. GOALS

GOAL 1: Gather accurate and accessible data on volume, turning movements, and traveler types at selected locations:

- Volume counts of people passing through pilot project areas
- Volumes accounted for by mode
- Desire lines and movement patterns accounted for by mode
- Accuracy of data at various locations and under a diversity of weather conditions/ times of day based on comparison with in-person traffic counts and historical data.

GOAL 2: Gather useful, usable data on street-level activity using sensors without contributing to the surveillance of residents:

- Collect minimum data required to generate actionable insights for planning
- Make data collection more accessible, trustworthy, and useful to the public.

II. PARTICIPATING PARTIES

Numina (a company that specializes in privacy-first multimodal data collection methodologies — more info at numina.co)

- Provide sensor-based data collection and reporting.

Mayor's Office of New Urban Mechanics (MONUM)

- Coordinate across departments and communities to explore, experiment, and evaluate this new approach to streets data collection.

Boston Transportation Department (BTD)

- Whose mission is to plan, build, maintain, and operate our streets in a manner that increases access to opportunities, promotes safety and resilience, and fosters the highest quality of life for everyone living, working, and visiting our City.
- Install sensors and provide feedback on use cases for data.

Analytics Team (Department of Innovation and Technology)

- Provide feedback on insights generated and coordinate data sharing and publication.

III. DATES

July 7, 2020 - Feb 11 2021